

Muhammad Ibrahim

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EDUCATION

New York University <i>Bachelor of Science in General Engineering, Mechanical and Civil Focus</i> - Major GPA - 3.60 / 4.00, UK (2:1). - Major Coursework - Solid Mechanics, Engineering Dynamics, Fluid Mechanics, Heat Transport, Composite Materials.	Abu Dhabi, UAE May 2026
New York University <i>Study Away, Immigration and Theatre</i> Analysed migration in French theatre casting and the economics of integration; Term GPA - 3.70 / 4.00.	Paris, France Jan 2024
New York University June Term <i>Study Away, Architecture and Climate Impact</i> Examined how architectural decisions shape climate outcomes and purpose-led design; Term GPA - 4.00 / 4.00.	Valletta, Malta Jun 2023

EXPERIENCE

Center for Stability, Instability and Turbulence (SITE) <i>Graduate Intern, New York University</i> - Delivered the model's entire training dataset solo in 6 weeks, 60 trials across 4 angles of attack. - Tripled the model's valid range to 45 degrees from a 15 degree dataset, grounding it in dimensional analysis. - Eliminated 50+ further tunnel runs, with a model trained in Python and MATLAB to predict untested angles.	New York, USA May 2026 - Jul 2026
Advanced Manufacturing Workshop <i>Workshop Technician, New York University</i> - Cut print failure rates from 50% to 10% across over 70 print hours by diagnosing root causes and refining parameters. - Raised part strength 50% over plain printed filament, laying up bidirectional carbon fibre and fibreglass.	Abu Dhabi, UAE Sep 2025 - May 2026
Center for Stability, Instability and Turbulence (SITE) <i>Undergraduate Research Assistant, New York University</i> - Generated a 130+ case Rayleigh-Taylor and Richtmyer-Meshkov dataset in a 2-person team, now under post-processing. - Achieved boundary-layer separation in every run across 5 water-tunnel model variants, verified by flow visualisation.	Abu Dhabi, UAE Apr 2023 - May 2026
Energy and Propulsion Lab <i>Undergraduate Research Assistant, New York University</i> - Boosted upper-level ventilation 28% for AeroFarms growing towers, simulating airflow in OpenFOAM and Converge. - Won Hyundai NGV industry recognition and a grant, presenting Urban Air Mobility propulsion findings to a 25-member panel.	Abu Dhabi, UAE Nov 2022 - Nov 2023

PROJECTS

Fixed-Wing UAV for Search and Rescue Payload Deployment, Personal Project - Drove a 1,130 mm 3D-printed airframe and 6-part powertrain from concept to finished prototype. - Reduced airframe mass from 2500 to 1500 g via lightweight PLA, infill tuning, internal geometry, and carbon tubing. - Engineered an airborne release for a 600 g multirotor daughter craft, delivered as a prototype for flight testing.	Jun 2026 - Present
Advanced Anti-Roll Tank System (ART3S), Mubadala Investment and Abu Dhabi Maritime Academy - Cut vessel roll amplitude 85% passively, engineering 200 to 800 m³ anti-heeling tanks for ferry and car-carrier hulls. - Modelled vessel roll decay in Simulink, cutting decay time 33% with passive tanks and 62% with active control.	Oct 2024 - Sep 2025
Savonius Turbine for Traffic Wind Energy Harvesting, Capstone Design - Raised peak moment 71.65% over the plain baseline, adding serrated edges and a 45-degree helical twist in ANSYS Fluent. - Delivered a 56.33% power increase over the non-serrated baseline in CFD, by adding serrated leading edges. - Proved roadside self-start within 2 to 5 seconds on a 1:3 carbon fibre rotor, projecting 3 to 15 W at full scale.	Sep 2024 - May 2025
Compressor-Less Multi-Stage TEC Refrigerator, New York University - Replaced compressor and refrigerant with 5 stacked thermoelectric modules, cooling 27 to 10 degrees in 24 minutes. - Tuned a closed-loop PID controller on Arduino, holding the 10-degree setpoint within the ±0.5-degree sensor accuracy.	Mar 2025 - May 2025
Formula Student SAE Drivetrain Design, New York University - Exposed a half-shaft 49% over its twist limit before build, by simulating shock loads that static checks had missed. - Redirected the redesign to geometry, saving the cost of an alloy with zero twist gain, by proving stiffness ignores yield.	Sep 2024 - May 2025

AWARDS

Research Grant, Mubadala and Abu Dhabi Maritime Association	Aug 2024
Industry Recognition and Grant, Hyundai NGV	Oct 2022
Full Merit Scholarship, New York University	Dec 2021

EXTRACURRICULAR ACTIVITIES

NYU Abu Dhabi Athletics, Varsity Jiu-Jitsu, Recreational Sport, and Team Videographer - Competed in jiu-jitsu at university varsity level, earning selection in each of four years. - Filmed and photographed the university cricket team across a full season, delivering footage and stills each week. - Recreational sport across padel, golf, tennis, cricket, volleyball, badminton, football, cycling, and wall climbing.	2022 - 2026
Habitat for Humanity Nepal, Volunteer, Residential Build Programme - Constructed a single-storey house from foundation to roof with a volunteer team, handing a finished home to its family. - Prevented build errors before concrete was poured, checking site dimensions and layouts against the civil drawings. - Converted donated labour and materials into permanent housing for families priced out of building.	2025
Emirates Motorsports Organization, Chief Track Marshal Directed trackside safety and first response across 15 events at Yas Marina Circuit and Dubai Autodrome.	2019 - 2025

SKILLS

CAD and Design - SolidWorks (CSWA), mechanism design, design for manufacturing, rapid prototyping.
Simulation and Analysis - CFD, finite element analysis (FEA), ANSYS Fluent, OpenFOAM, Converge, Simulink, HPC.
Manufacturing - FDM 3D printing, carbon fibre and fibreglass composites, Arduino, closed-loop PID control.
Programming - Python, MATLAB, machine learning, sensor data acquisition.
Languages - English C2, Urdu C2, Arabic A2.